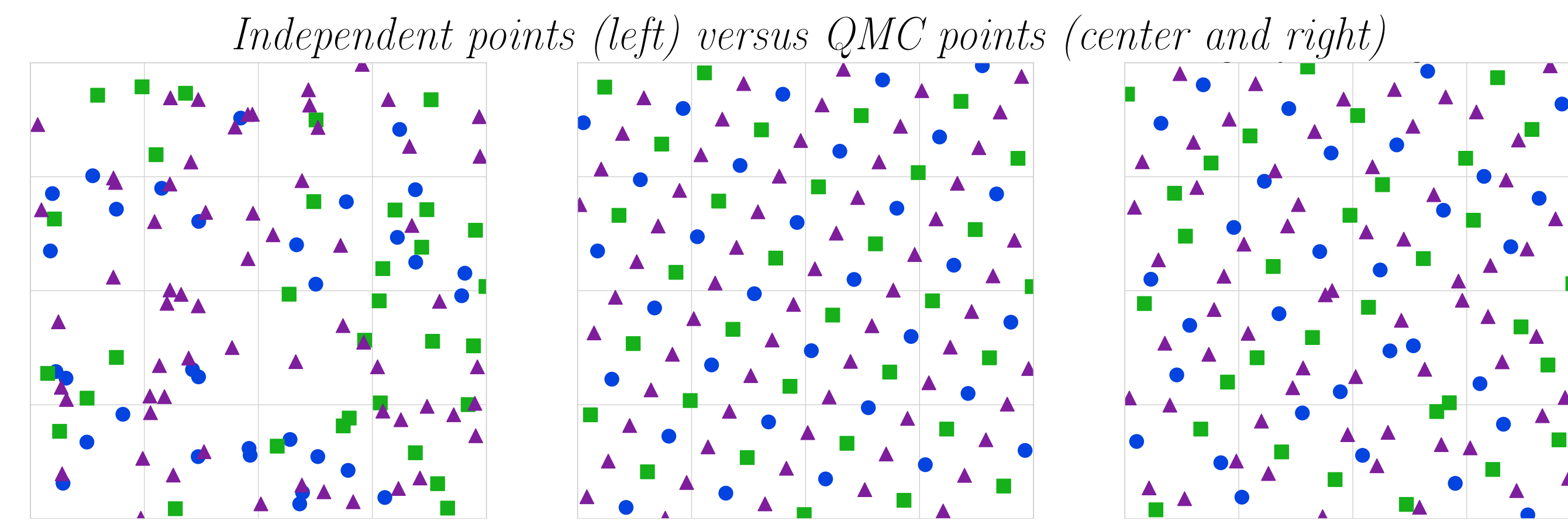


High-dimensional numerical integration: Monte Carlo and quasi-Monte Carlo (QMC)

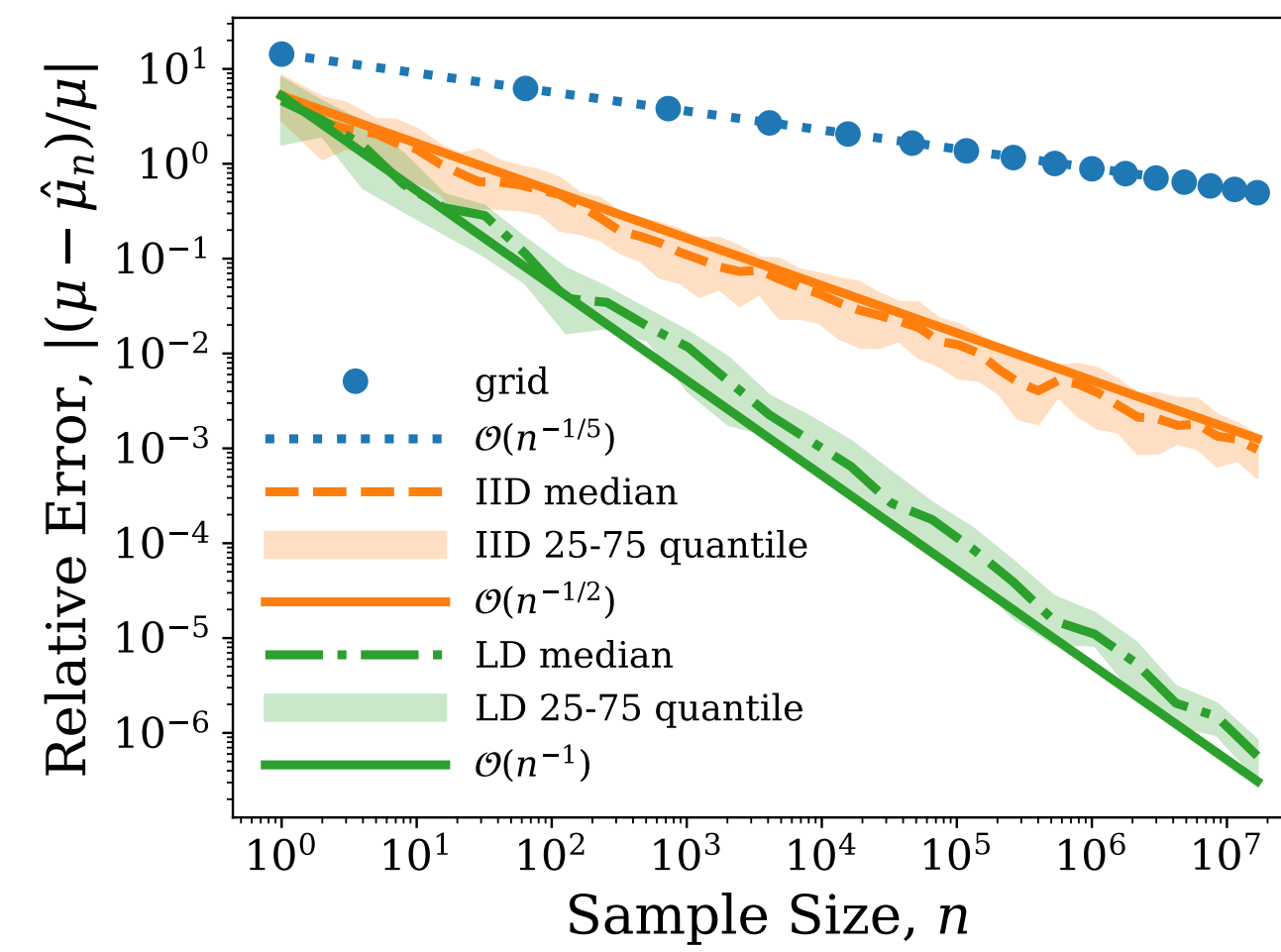
- High-dimension numerical integration is crucial across many scientific disciplines e.g. financial modeling, solving differential equations, graphics rendering, ...

$$\int_{[0,1]^d} f(\mathbf{x}) d\mathbf{x} \approx \frac{1}{N} \sum_{i=1}^N f(\mathbf{x}_i), \quad \mathbf{x}_1, \dots, \mathbf{x}_n \in [0,1]^d$$

- Grid and sparse grid points $\{\mathbf{x}_i\}_{i=1}^n$ are inefficient in high dimensions
- Monte Carlo uses independent points $\{\mathbf{x}_i\}_{i=1}^n$: error like $\mathcal{O}(N^{-1/2})$
- Quasi-Monte Carlo uses carefully coordinated points $\{\mathbf{x}_i\}_{i=1}^n$: errors $\approx \mathcal{O}(N^{-1})$



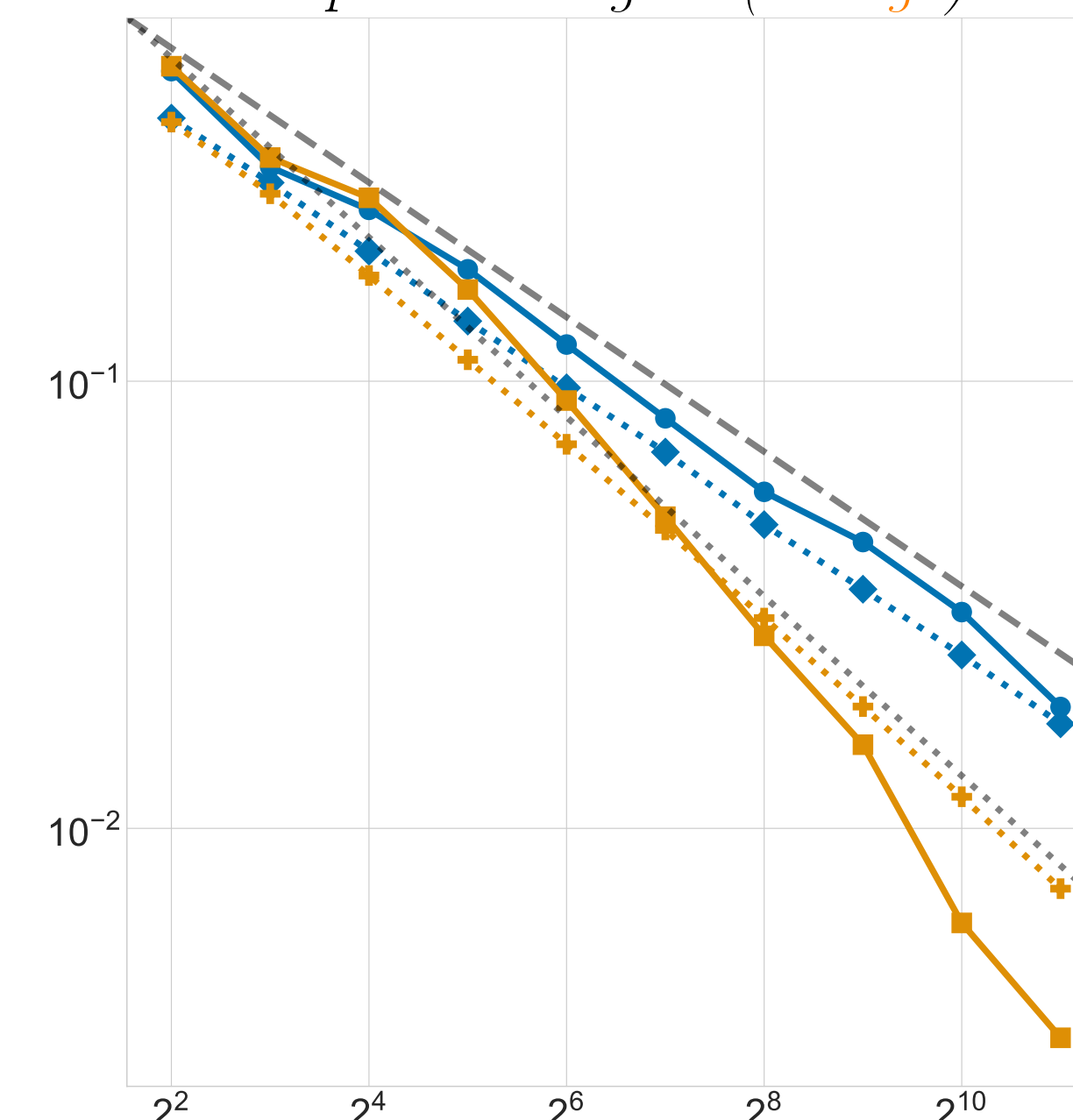
- The greater uniformity of QMC points leads to faster convergence.
- Developed QMCPy [14, 4, 13, 2, 3, 6] qmcssoftware.github.io/QMCSsoftware/ with low-level generators in C qmcssoftware.github.io/QMCToolsCL/ backed by our QMC data repository github.com/QMCSsoftware/LDDData



Optimal Monte Carlo on matrix manifolds

- Integration over matrix manifolds is used in medical imaging, materials science, ... to characterize random low-dimensional projections of high-dimensional shapes
- Consider smooth functions (Sobolev- s spaces) f of random projections $\{\mathbf{x}_i\}_{i=1}^N$
- Consider a weighted approximation $\sum_{i=1}^N w_i f(\mathbf{x}_i)$ with weights $\{w_i\}_{i=1}^N$

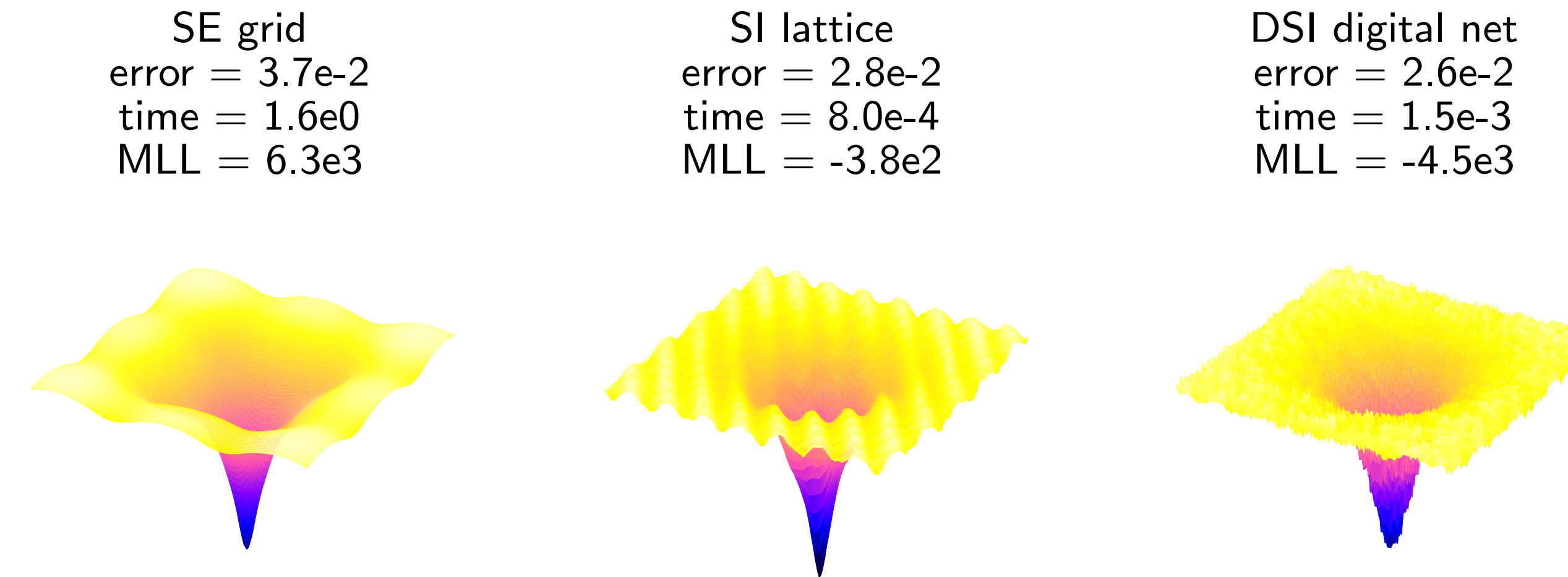
Error when using equal weights (blue)
versus optimal weights (orange)



- Classic Monte Carlo with independent points $\{\mathbf{x}_i\}_{i=1}^N$ and equal weights $w_i = 1/N$ still has error like $\mathcal{O}(N^{-1/2})$
- Independent points $\{\mathbf{x}_i\}_{i=1}^N$ and optimized weights $\{w_i\}_{i=1}^N$ achieve optimal error rate $\mathcal{O}(N^{-s/d}) \lesssim \mathcal{O}(N^{-1/2})$, $s > d/2$.
- Optimization of weights costs $\mathcal{O}(N^3)$ but enables optimal rate while still using independent points.
- Application to simultaneous intrinsic volume estimation using flag manifold generalization of Grassmann manifolds

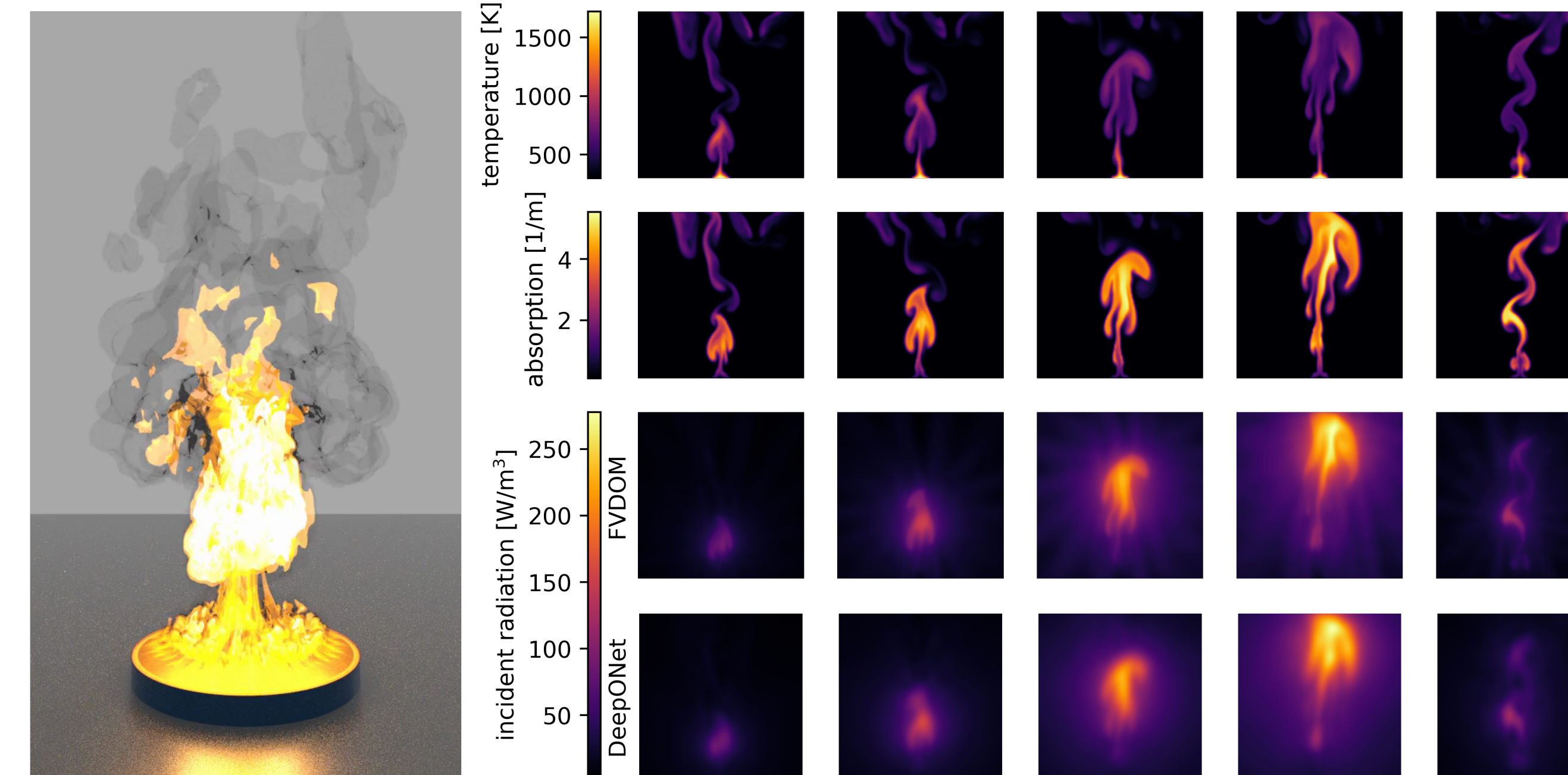
Fast Gaussian process (GP) regression probability models

- GP regression models provide predictions with built-in uncertainty quantification
- Capable of encoding prior information via the covariance kernel specification
- Integrals of GPs are Gaussian, connections with (Q)MC
- Standard GP regression requires $\mathcal{O}(N^3)$ computations and $\mathcal{O}(N^2)$ storage
- Pairing QMC points with special covariance kernels incudes nice structure $\Rightarrow$ "fast GPs" only requires $\mathcal{O}(N \log N)$ computations and $\mathcal{O}(N)$ storage.
- Generalized structural exploits to fast multitask GPs [11, 8, 10, 12]
- Developed **fastgps** Python software: alegresor.github.io/fastgps/

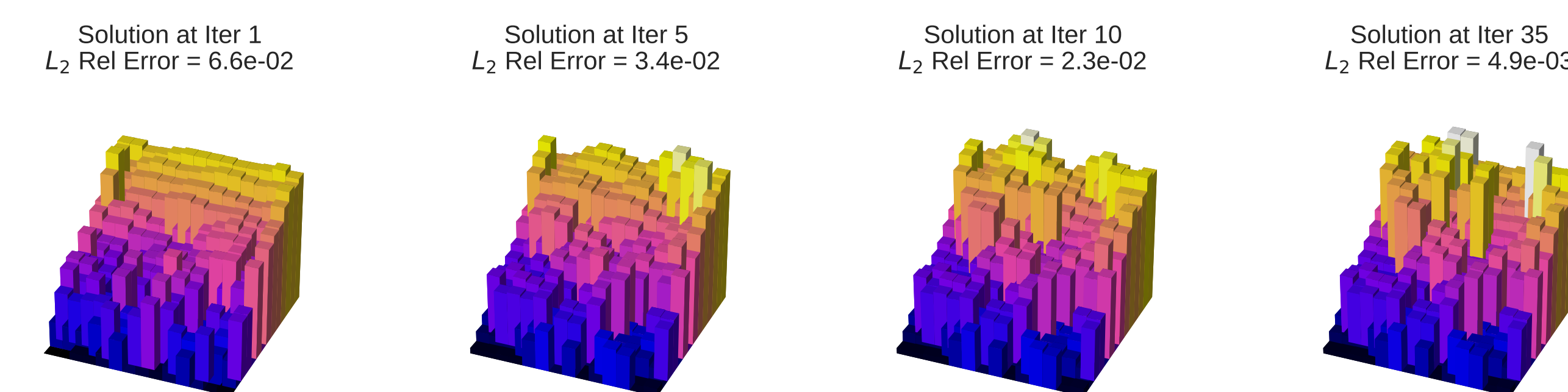


Solving nonlinear PDEs with random coefficients

- Nonlinear PDEs (partial differential equations) govern many physical phenomena, e.g., reaction-diffusion states, groundwater flow, acoustic imaging, ...
- Explored scientific machine learning for fire simulations via radiative transfer [7]



- Solution v of nonlinear PDE F with coefficients c via successive linearization
- $$v_{n+1}(c) = v_n(c) - [J(v, c)^* J(v, c) + \lambda I]^{-1} J(v, c)^* F(v, c), \quad J(v, c) = \frac{\delta F}{\delta v}(v, c)$$



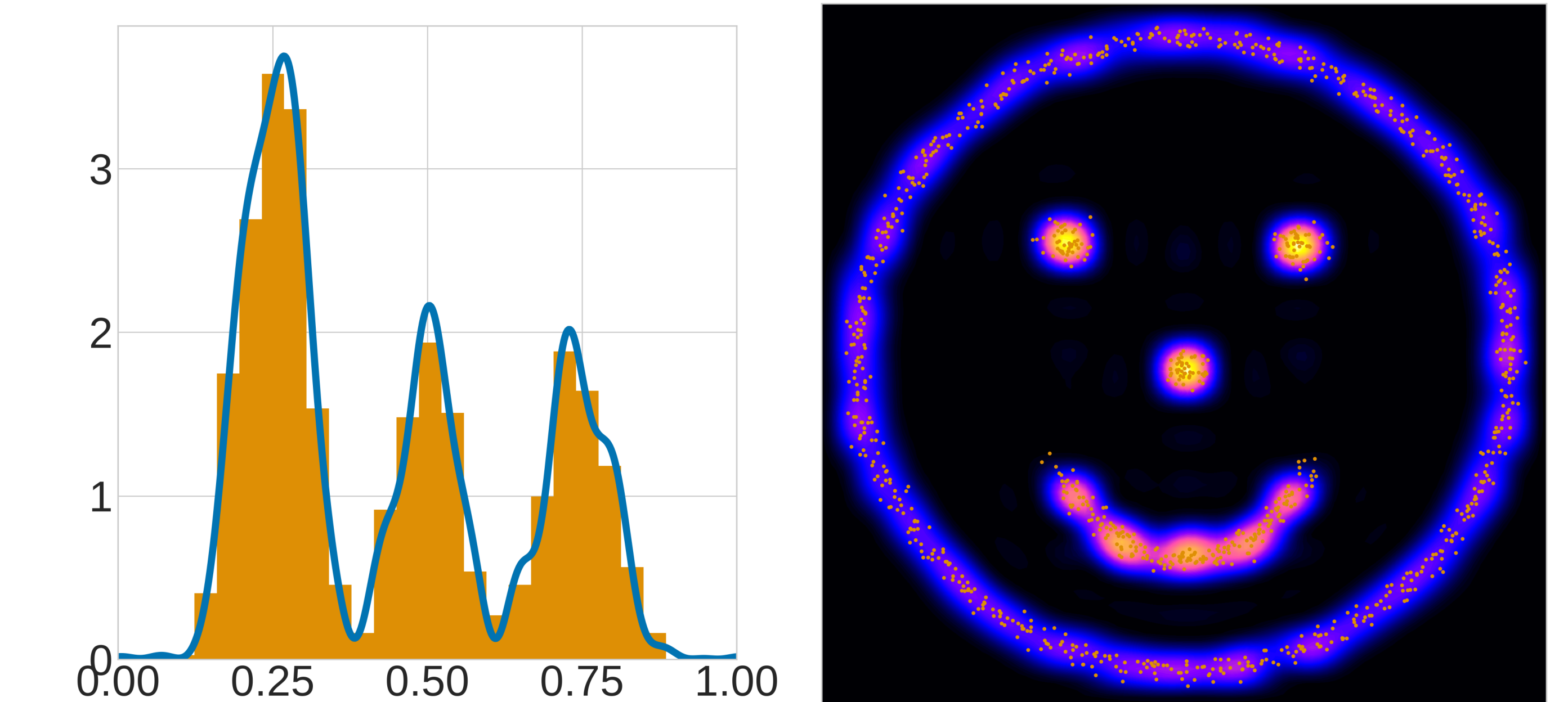
- Enabled machine learning of $[J(v, c)^* J(v, c) + \lambda I]$ via Cholesky factors [1]
- Developed scalable software github.com/OwhadiGroup/CHONKNORIS

Parameteric density estimation via convex optimization

- Density estimation has proven especially useful in physical sciences, e.g., modeling animal clusters, matter distributions, disease locations, ...
- Want analytic and non-negative density which integrates to one

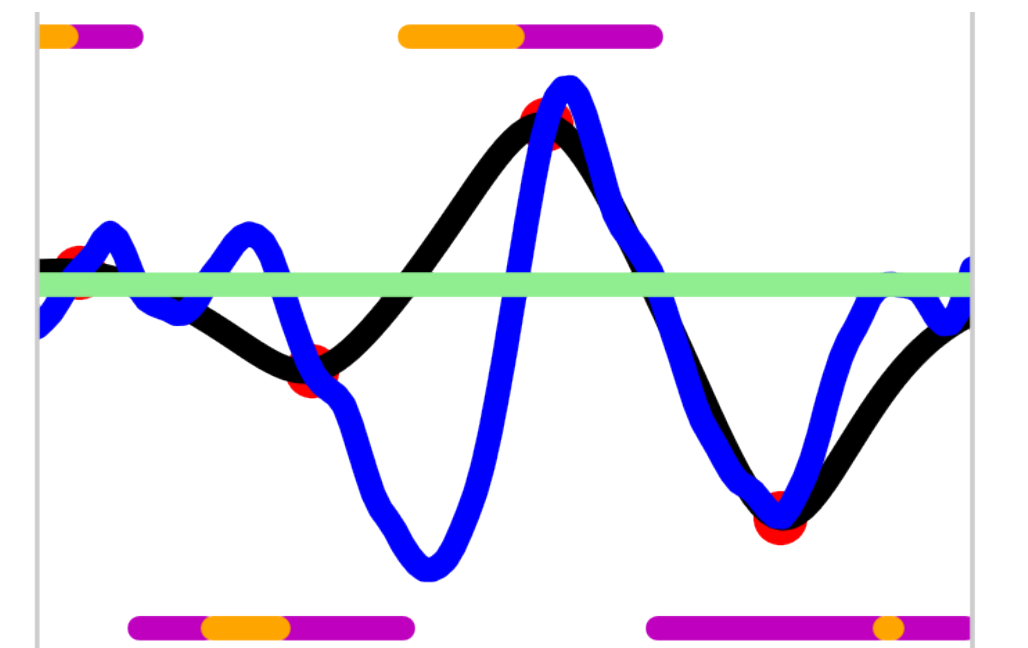
$$\rho(\mathbf{x}) = \sum_{i,j=1}^T \overline{G_{ij}} \psi_i(\mathbf{x}) \overline{\psi_j(\mathbf{x})} = \psi(\mathbf{x})^\dagger \mathbf{G} \psi(\mathbf{x}), \quad \mathbf{G} \succeq 0$$

- Solving for positive semi-definite (PSD) $\mathbf{G}$ is a convex optimization problem



Error-aware reliability analysis with GPs

- Reliability analysis used in rare event modeling, e.g., extreme weather, system diagnostics, ...
- Derived GP credible intervals on the probability of failure and proposed a new Bayesian-opt sample-acquisition scheme [9]



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